# Jacobian Descent For Multi-Objective Optimization

Valérian Rey — Simplex Lab
valerian.rey@gmail.com
Joint work with Pierre Quinton

February 11, 2026

GitHub
github.com/TorchJD/torchjd

TorchJD
torchjd.org

arXiv
arxiv.org/pdf/2406.16232

# Table of contents

# Theory

# Partial order

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m$, as

$$\begin{aligned} & \boldsymbol{u} \preceq \boldsymbol{v} & \\ \Leftrightarrow \quad & u_i \leq v_i & \forall i \in [m] \end{aligned}$$

# Partial order

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m$, as

$$\begin{aligned} & \boldsymbol{u} \preceq \boldsymbol{v} & \\ \Leftrightarrow \qquad & u_i \leq v_i & \forall i \in [m] \end{aligned}$$

Examples:

- $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \preceq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

# Partial order

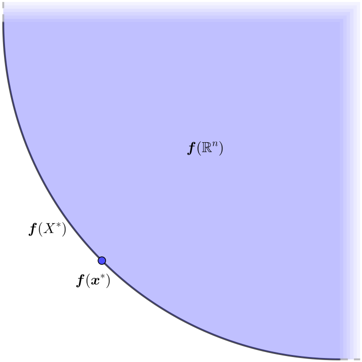
Define the partial order $(\mathbb{R}^m, \preceq)$, for $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m$, as

$$\begin{aligned} & \boldsymbol{u} \preceq \boldsymbol{v} \\ \Leftrightarrow \quad & u_i \leq v_i \qquad \forall i \in [m] \end{aligned}$$

Examples:

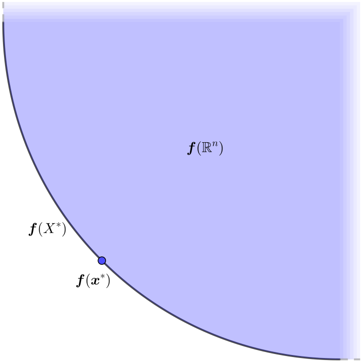
- $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \preceq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$
- $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are not comparable with this partial order.

# Pareto optimality



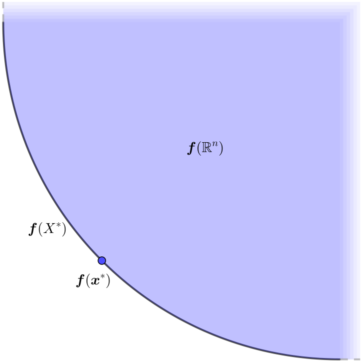
# Pareto optimality

Pareto set: $X^*$



# Pareto optimality

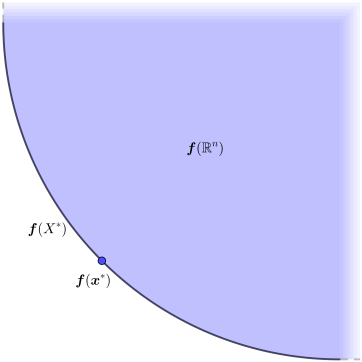
Pareto set: $X^*$
Pareto Front: $\boldsymbol{f}(X^*)$



# Pareto optimality

Pareto set: $X^*$

Pareto Front: $\boldsymbol{f}(X^*)$




$$\boldsymbol{x}^* \in X^* \quad \Leftrightarrow \quad \forall \boldsymbol{y} \in \mathbb{R}^n,\ \boldsymbol{f}(\boldsymbol{y}) \preceq \boldsymbol{f}(\boldsymbol{x}^*) \rightarrow \boldsymbol{f}(\boldsymbol{y}) = \boldsymbol{f}(\boldsymbol{x}^*)$$

# Jacobian descent

Goal: minimize $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ according to the $\preceq$ partial order.

# Jacobian descent

Goal: minimize $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ according to the $\preceq$ partial order. For any $\boldsymbol{x} \in \mathbb{R}^n$, the Jacobian of $\boldsymbol{f}$ at $\boldsymbol{x}$, is

$$\mathcal{J}\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} \nabla f_1(\boldsymbol{x})^\top \\ \nabla f_2(\boldsymbol{x})^\top \\ \vdots \\ \nabla f_m(\boldsymbol{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_1(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\boldsymbol{x}) \\ \frac{\partial}{\partial x_1} f_2(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_2(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\boldsymbol{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_m(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\boldsymbol{x}) \end{bmatrix}$$

# Jacobian descent

Goal: minimize $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ according to the $\preceq$ partial order. For any $\boldsymbol{x} \in \mathbb{R}^n$, the Jacobian of $\boldsymbol{f}$ at $\boldsymbol{x}$, is

$$\mathcal{J}\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} \nabla f_1(\boldsymbol{x})^\top \\ \nabla f_2(\boldsymbol{x})^\top \\ \vdots \\ \nabla f_m(\boldsymbol{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_1(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\boldsymbol{x}) \\ \frac{\partial}{\partial x_1} f_2(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_2(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\boldsymbol{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_m(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\boldsymbol{x}) \end{bmatrix}$$

For a small enough update $\boldsymbol{y} \in \mathbb{R}^n$, Taylor's theorem states

$$\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \approx \boldsymbol{f}(\boldsymbol{x}) + \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{y}$$

# Jacobian descent

Goal: minimize $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ according to the $\preceq$ partial order. For any $\boldsymbol{x} \in \mathbb{R}^n$, the Jacobian of $\boldsymbol{f}$ at $\boldsymbol{x}$, is

$$\mathcal{J}\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} \nabla f_1(\boldsymbol{x})^\top \\ \nabla f_2(\boldsymbol{x})^\top \\ \vdots \\ \nabla f_m(\boldsymbol{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_1(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\boldsymbol{x}) \\ \frac{\partial}{\partial x_1} f_2(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_2(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\boldsymbol{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_m(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\boldsymbol{x}) \end{bmatrix}$$

For a small enough update $\boldsymbol{y} \in \mathbb{R}^n$, Taylor's theorem states

$$\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \approx \boldsymbol{f}(\boldsymbol{x}) + \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{y}$$

We want $\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \preceq \boldsymbol{f}(\boldsymbol{x})$.

# Jacobian descent

Goal: minimize $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ according to the $\preceq$ partial order. For any $\boldsymbol{x} \in \mathbb{R}^n$, the Jacobian of $\boldsymbol{f}$ at $\boldsymbol{x}$, is

$$\mathcal{J}\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} \nabla f_1(\boldsymbol{x})^\top \\ \nabla f_2(\boldsymbol{x})^\top \\ \vdots \\ \nabla f_m(\boldsymbol{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_1(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\boldsymbol{x}) \\ \frac{\partial}{\partial x_1} f_2(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_2(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\boldsymbol{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_m(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\boldsymbol{x}) \end{bmatrix}$$

For a small enough update $\boldsymbol{y} \in \mathbb{R}^n$, Taylor's theorem states

$$\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \approx \boldsymbol{f}(\boldsymbol{x}) + \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{y}$$

We want $\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \preceq \boldsymbol{f}(\boldsymbol{x})$. We select $\boldsymbol{y} = -\eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x})) \in \mathbb{R}^n$.

# Jacobian descent

The mapping $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$ is called an **aggregator**, it parameterizes **Jacobian descent**:

# Jacobian descent

The mapping $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$ is called an **aggregator**, it parameterizes **Jacobian descent**:

---

**Algorithm 1:** Jacobian descent with aggregator $\mathcal{A}$

---

**Input:** $\boldsymbol{x} \in \mathbb{R}^n$, $0 < \eta$, $T \in \mathbb{N}$, $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$

**for** $t \leftarrow 1$ **to** $T$ **do**

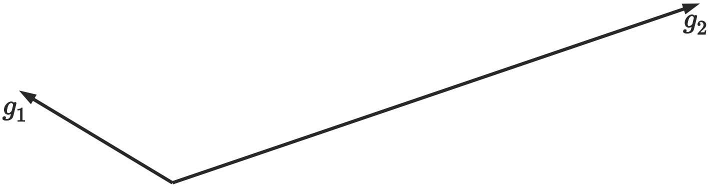
  $\boldsymbol{x} \leftarrow \boldsymbol{x} - \eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x}))$

**Output:** $\boldsymbol{x}$

---

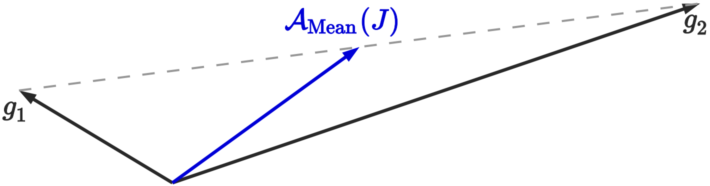
# Unconflicting projection of gradients (UPGrad)

$$J = \begin{bmatrix} g_1^\top \\ g_2^\top \end{bmatrix}$$



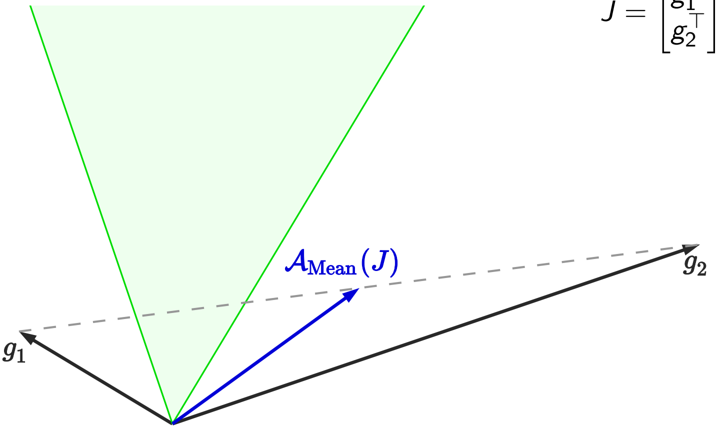
# Unconflicting projection of gradients (UPGrad)

$$J = \begin{bmatrix} g_1^\top \\ g_2^\top \end{bmatrix}$$

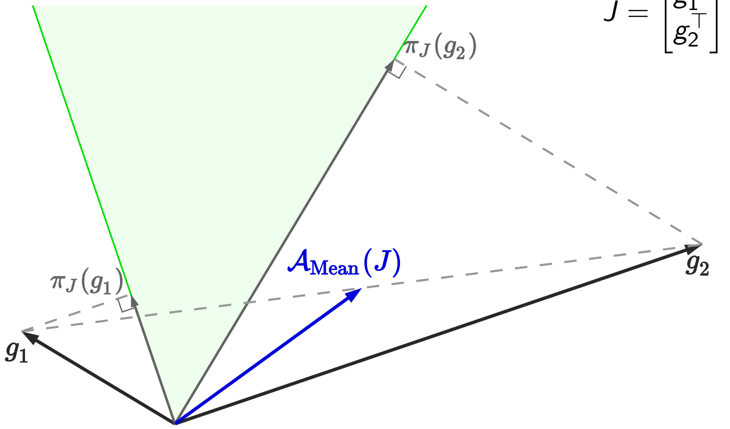


# Unconflicting projection of gradients (UPGrad)

$$J = \begin{bmatrix} g_1^\top \\ g_2^\top \end{bmatrix}$$

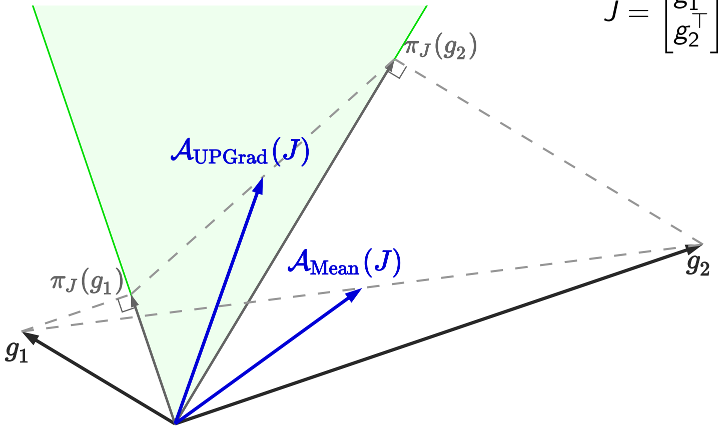


# Unconflicting projection of gradients (UPGrad)




$$J = \begin{bmatrix} g_1^\top \\ g_2^\top \end{bmatrix}$$

# Unconflicting projection of gradients (UPGrad)




$$J = \begin{bmatrix} g_1^\top \\ g_2^\top \end{bmatrix}$$

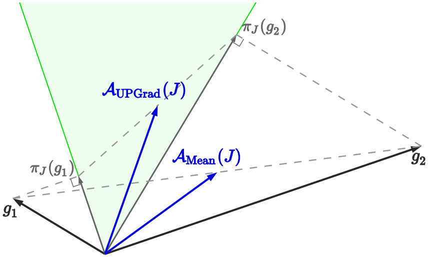
# Formal definition of UPGrad

Given $J \in \mathbb{R}^{m \times n}$, let $C_J$ be the cone $\{\boldsymbol{y} \in \mathbb{R}^n : \mathbf{0} \preceq J\boldsymbol{y}\}$. $\mathcal{A}$ is non-conflicting if for all $J \in \mathbb{R}^{m \times n}$, $\mathcal{A}(J) \in C_J$.

For any $\boldsymbol{x} \in \mathbb{R}^n$, the projection of $\boldsymbol{x}$ onto $C_J$ is

$$\pi_J(\boldsymbol{x}) = \underset{\boldsymbol{y} \in C_J}{\arg\min} \|\boldsymbol{x} - \boldsymbol{y}\|^2$$

$$\mathcal{A}_{\text{UPGrad}}(J) = \frac{1}{m} \sum_{i \in [m]} \pi_J(g_1)$$



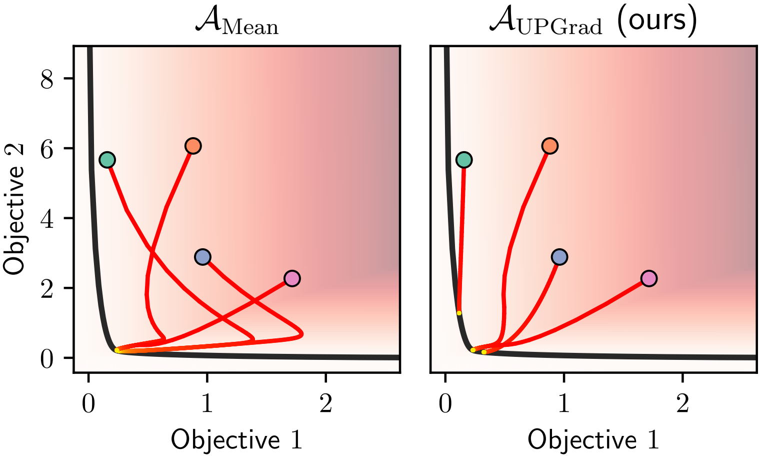
# Applications and experimentation

# Optimization trajectories

Goal: Minimize some convex quadratic function $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^2$, with conflicting gradients.

# Optimization trajectories

Goal: Minimize some convex quadratic function $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^2$, with conflicting gradients.



# Optimization trajectories: code

```
x = tensor([0., 0.])  # initial value of x

for i in range(n_iters):
    y = f(x)  # shape: [2]
    autojac.backward(y)
    # x now has a .jac field
    autojac.jac_to_grad([x], UPGrad())
    # x now has a .grad field
    optimizer.step()
    optimizer.zero_grad()
```

Full project available at github.com/TorchJD/trajectories

# Per-sample JD

Each element of the batch has its own loss.

# Per-sample JD

Each element of the batch has its own loss.

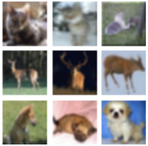
```
batch_size = 16
model = Linear(10, 1)

for x, y in dataloader:
  z = model(x).squeeze(dim=1)  # shape: [16]
  losses = loss_fn(z, y)  # shape: [16]
  autojac.backward(losses)
  # parameters now have a .jac field
  autojac.jac_to_grad(model.parameters(), UPGrad())
  # parameters now have a .grad field
  optimizer.step()
  optimizer.zero_grad()
```

Full code example available at torchjd.org/latest/examples/iwrm.

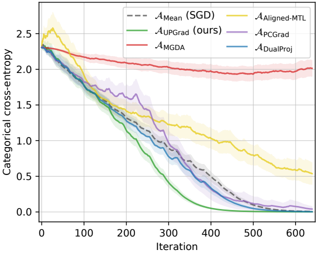
# Per-sample JD: results

Batch size: 32

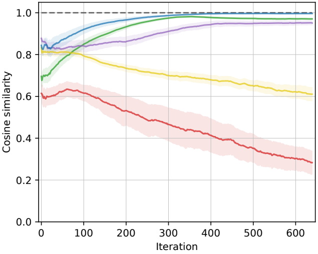
CIFAR-10



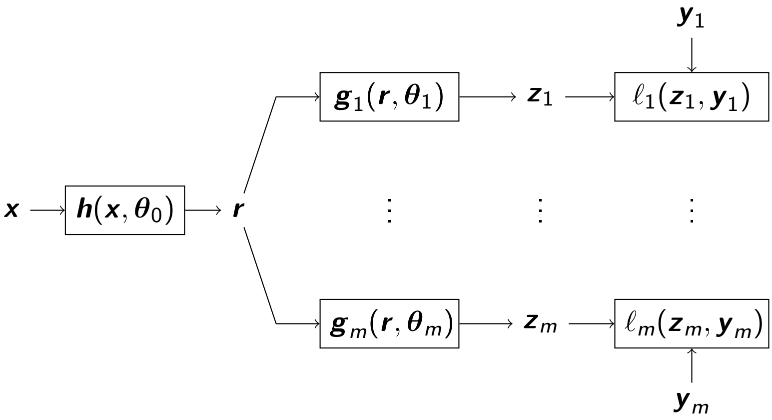
Training Loss



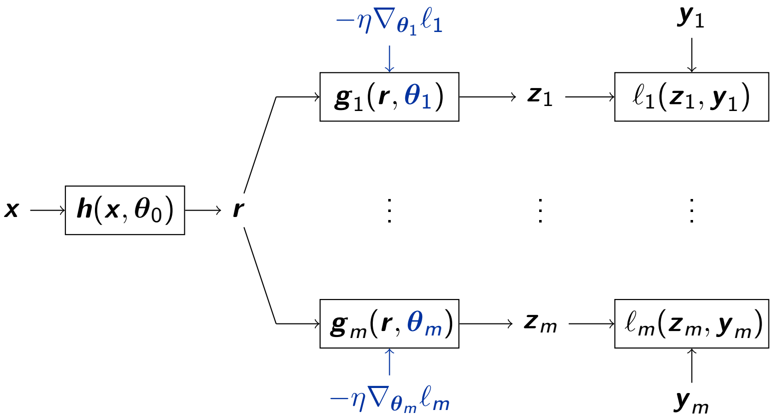

Update Similarity



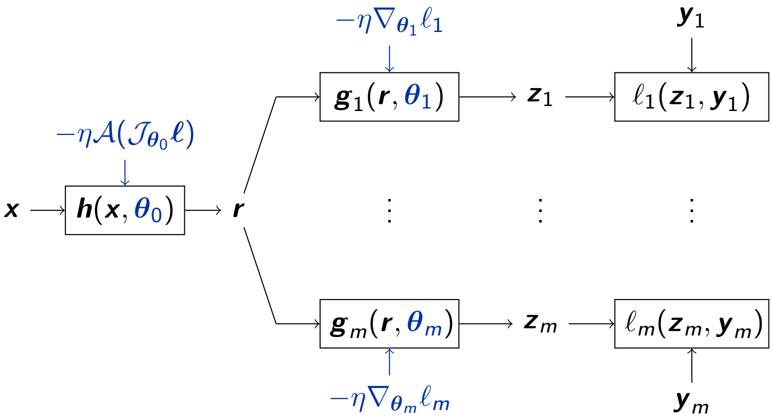
# Multi-task learning



# Multi-task learning



# Multi-task learning



# Multi-task learning: code

```python
h = Linear(10, 5)
g1 = Linear(5, 1)
g2 = Linear(5, 1)

for x, y1, y2 in dataloader:
  r = h(x)
  z1 = g1(r)
  z2 = g2(r)
  l1 = loss_fn(z1, y1)
  l2 = loss_fn(z2, y2)
  autojac.mtl_backward([l1, l2], features=r)
  # head parameters now have a .grad field
  # backbone parameters now have a .jac field
  autojac.jac_to_grad(h.parameters(), UPGrad())
  # All parameters now have a .grad field
  optimizer.step()
  optimizer.zero_grad()
```

Full code example available at torchjd.org/latest/examples/mtl.

# Conclusion

Many problems are multi-objective.

# Conclusion

Many problems are multi-objective.

Dual cone is rarely empty in deep learning.

# Conclusion

Many problems are multi-objective.

Dual cone is rarely empty in deep learning.

TorchJD to compute and aggregate Jacobians.

# Conclusion

Many problems are multi-objective.

Dual cone is rarely empty in deep learning.

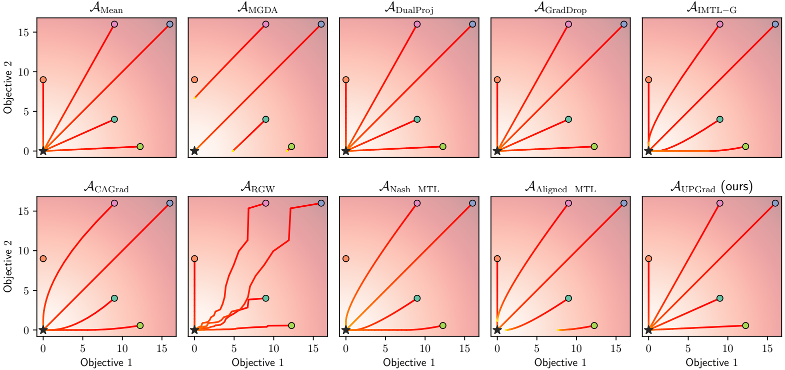
TorchJD to compute and aggregate Jacobians.

Try to compute the inner products between your gradients.

# Appendix

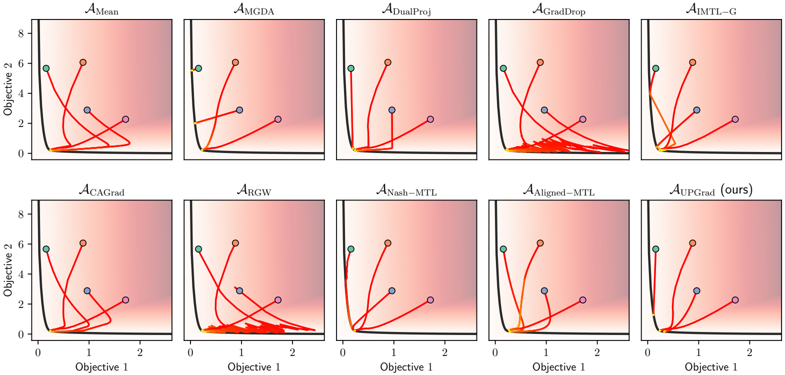
# Element-wise quadratic function trajectories

$$\text{Minimize } \boldsymbol{f} : \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x^2 \\ y^2 \end{bmatrix}. \text{ It's Jacobian at } \begin{bmatrix} x \\ y \end{bmatrix} \text{ is } \begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}.$$



# Convex quadratic form trajectories

Some convex quadratic function $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^2$, with conflicting gradients.



# Gramian-based JD

$\mathcal{A}_{\text{UPGrad}}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x}))$ is a linear combination of the rows of $\mathcal{J}\boldsymbol{f}(\boldsymbol{x})$.

# Gramian-based JD

$\mathcal{A}_{\text{UPGrad}}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x}))$ is a linear combination of the rows of $\mathcal{J}\boldsymbol{f}(\boldsymbol{x})$.

The weights only depend on the (small) Gramian $\mathcal{G}\boldsymbol{f}(\boldsymbol{x}) = \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \mathcal{J}\boldsymbol{f}(\boldsymbol{x})^\top$.

## Gramian-based JD

$\mathcal{A}_{\mathsf{UPGrad}}\big(\mathcal{J}\boldsymbol{f}(\boldsymbol{x})\big)$ is a linear combination of the rows of $\mathcal{J}\boldsymbol{f}(\boldsymbol{x})$.

The weights only depend on the (small) Gramian $\mathcal{G}\boldsymbol{f}(\boldsymbol{x}) = \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \mathcal{J}\boldsymbol{f}(\boldsymbol{x})^\top$.

We can compute $\mathcal{G}\boldsymbol{f}(\boldsymbol{x})$ directly, extract the weights, combine the losses, and do a step of gradient descent.

## Gramian-based JD

$\mathcal{A}_{\text{UPGrad}}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x}))$ is a linear combination of the rows of $\mathcal{J}\boldsymbol{f}(\boldsymbol{x})$.

The weights only depend on the (small) Gramian $\mathcal{G}\boldsymbol{f}(\boldsymbol{x}) = \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \mathcal{J}\boldsymbol{f}(\boldsymbol{x})^\top$.

We can compute $\mathcal{G}\boldsymbol{f}(\boldsymbol{x})$ directly, extract the weights, combine the losses, and do a step of gradient descent.

We have developed the `autogram` engine to compute $\mathcal{G}\boldsymbol{f}(\boldsymbol{x})$ efficiently (work in progress).

# Gramian-based JD

$\mathcal{A}_{\text{UPGrad}}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x}))$ is a linear combination of the rows of $\mathcal{J}\boldsymbol{f}(\boldsymbol{x})$.

The weights only depend on the (small) Gramian $\mathcal{G}\boldsymbol{f}(\boldsymbol{x}) = \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \mathcal{J}\boldsymbol{f}(\boldsymbol{x})^\top$.

We can compute $\mathcal{G}\boldsymbol{f}(\boldsymbol{x})$ directly, extract the weights, combine the losses, and do a step of gradient descent.

We have developed the `autogram` engine to compute $\mathcal{G}\boldsymbol{f}(\boldsymbol{x})$ efficiently (work in progress).

This saves a lot of memory, so it's often much faster.

# Gramian-based JD

$\mathcal{A}_{\text{UPGrad}}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x}))$ is a linear combination of the rows of $\mathcal{J}\boldsymbol{f}(\boldsymbol{x})$.

The weights only depend on the (small) Gramian $\mathcal{G}\boldsymbol{f}(\boldsymbol{x}) = \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \mathcal{J}\boldsymbol{f}(\boldsymbol{x})^\top$.

We can compute $\mathcal{G}\boldsymbol{f}(\boldsymbol{x})$ directly, extract the weights, combine the losses, and do a step of gradient descent.

We have developed the **`autogram`** engine to compute $\mathcal{G}\boldsymbol{f}(\boldsymbol{x})$ efficiently (work in progress).

This saves a lot of memory, so it's often much faster.

Applicable to most other aggregators from the literature.